LESSON PLAN

A/C MATERIAL (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	16	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Familiarization with various parts of aircraft and different arrangement of components as seen on various aircraft in station.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Familiarization with various parts of aircraft and different arrangement of components as seen on various aircraft in station.
5. Specific Objectives	15.1.1.1 Trainee will be able to explain aircraft structure, various structural elements and their details.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Identification of primary and secondary structural elements on a cutaway model; Safety: Handle models with care to avoid damage; Supervision: Instructor verifies correct identification of fuselage, wings, and tail surfaces.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the fuselage?
(b) Name two types of wings used in modern aircraft.
(c) Why is the empennage critical for flight stability?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Fuselage structure and types • Wing configuration and components • Empennage assembly details • Landing gear arrangement • Power plant location	L & Ds	PPT, IB	15 min
(b)	• Comparison of component layouts • Structural element identification • Material usage in structures • Component arrangement variations • Station-specific configurations	L & Ds	PPT, PDF	20 min
(c)	• Review of structural elements • Identification of key stations • Function of major components • Variations across aircraft types • Safety implications of structure	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between monocoque and semi-monocoque structures.
(b) How does the arrangement of landing gear affect ground handling?

12. Home Assignment and Follow Up	Prepare a brief note comparing the structural layout of the aircraft currently in station with one previously studied, focusing on component arrangement.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Overall consideration of aircraft loading action. Symmetric flight loads, symmetric maneuvers	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Wing Commander, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min